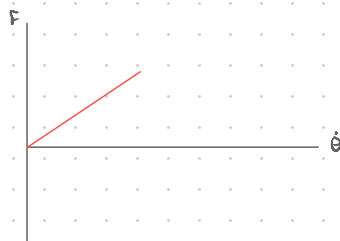
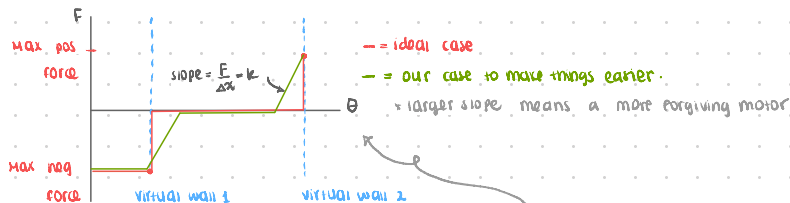
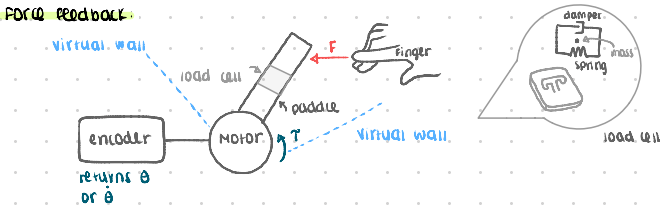


FORCE FEEDBACK



in PICO:

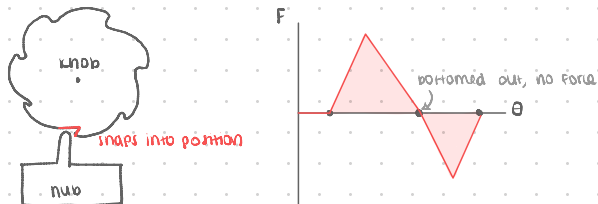
$$I = k_f e_f + k_d \dot{e}_f, \quad e_f = \text{desired force} - \text{actual force}$$

↑
error in the force

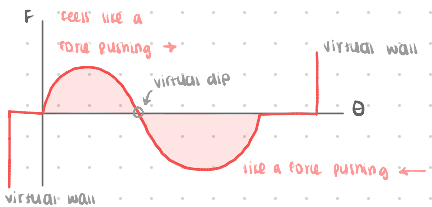
NOTE that $F = \cancel{k_d} \dot{\theta} + \cancel{k_d} \dot{\theta}$

$\cancel{\text{spring term}}$ damping term
 negligible

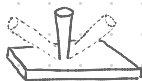
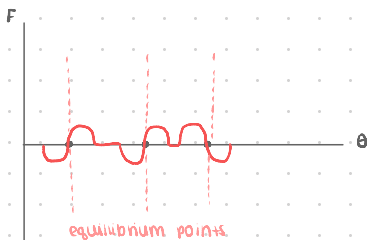
let's say we introduce a knob



to represent force as less discrete, we can use sine waves

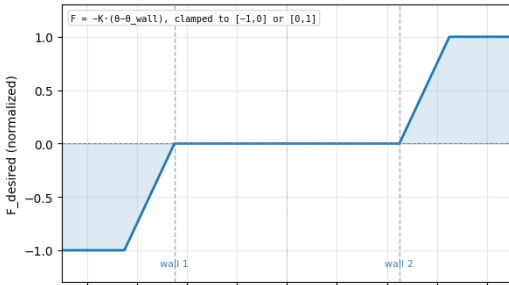


let's say we implement a toggle switch

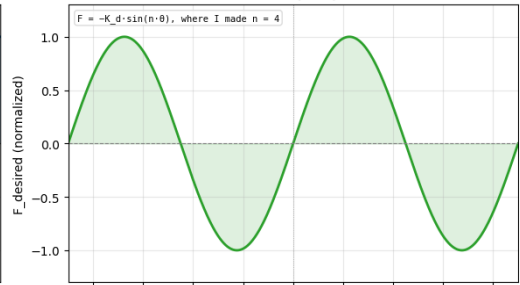


Haptic Paddle: Force vs. Displacement

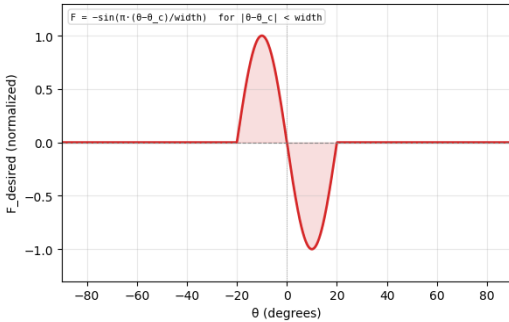
Virtual Walls



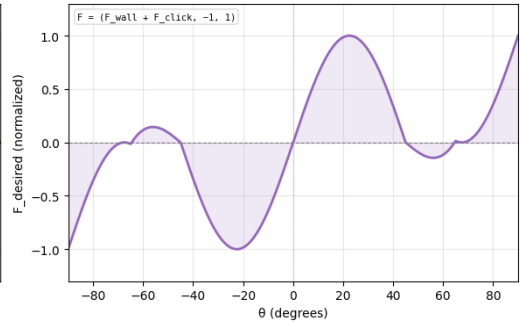
Clicks (n=4)



Bump / Dip



Full System



my haptic paddle visualizer

ME433 — Haptic Paddle Visualizer

$\theta = +25.0^\circ$

$F = -0.500$

Tab = cycle mode:

clicks

toggle

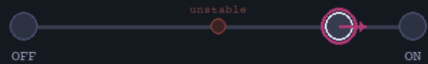
> bump



force vs θ [bump]



toggle switch (stable positions)



state: mid — unstable

drag : <- -> : Tab : Q